

NAMRATESH SHRIVASTAV

Senior AI Lead | Data Scientist

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PROFESSIONAL SUMMARY

Senior AI Lead with 7 years of expertise in architecting production-grade Generative AI systems and MLOps pipelines. Specialist in designing scalable RAG (Retrieval-Augmented Generation) architectures and reliability-first LLM middleware for enterprise-scale deployments. Proven track record in optimizing high-throughput AI systems with a deep mathematical foundation in deep learning and NLP.

TECHNICAL SKILLS

Generative AI & Agentic Systems: LangGraph (Multi-agent), LangChain, LlamaIndex, LangSmith (Evaluation), RAG Pipelines, Prompt Engineering, LLM API Orchestration (OpenAI/Claude/Gemini)

Machine Learning & Distributed Systems: PyTorch, TensorFlow, Scikit-Learn, Ray (Distributed Training/Scaling), Computer Vision (PaddleOCR, Tesseract), NLP, Model Optimization

Cloud Architecture (Multi-Cloud):

- **AWS:** SageMaker (End-to-End ML), Lambda (Serverless), API Gateway, ECR, Textract, Comprehend, Bedrock
- **Azure:** Azure AI Foundry, Logic Apps, Web App Services (PaaS), ACR, AKS, API Gate Way
- **GCP:** Vertex AI, Google Container Registry (GCR)

MLOps & Production Engineering: Docker, Kubernetes (AKS), MLFlow, DVC, GitHub Actions (CI/CD), Prometheus, Grafana, Jenkins, Redis, RabbitMQ/Celery (Asynchronous Task Queues)

Engineering & Data: Python, SQL, FastAPI, Streamlit, Gradio, Pymongo, Selenium, CI/CD, Agile, AI-Assisted Engineering (Claude Code, Cursor), Test-Driven Development (TDD)

PROFESSIONAL EXPERIENCE

Publicis Sapient

July 2024 – Present

RAG Solution with Azure OpenAI for Medical Policy Data

- Architected a production-grade Multi-Agent RAG System using Azure OpenAI and LlamaIndex to support multi-domain medical policy queries; successfully scaled and replicated the architecture for telesales agents, driving a 35% increase in agent efficiency.
- Pioneered a Custom AI Evaluation Framework with AI-assisted development workflows (Cursor, Claude Code) (extending the SAFE framework) that was adopted as the organizational default for all multi-domain releases, reducing regional GTM (Go-To-Market) cycles by 75% (from 4 months to 1 month).
- Engineered an Automated Transcript Analysis Pipeline utilizing Google Cloud Speech-to-Text to transcribe and analyze agent-customer interactions, enabling data-driven insights into sales performance and compliance.
- Operationalized a Robust Regression Testing Suite via Kubernetes CronJobs and GitHub Actions CI/CD; automated daily evaluation of 300+ complex medical scenarios to maintain 99.9% system reliability and zero-regression deployments.

Infrablok

Apr 2023 – June 2024

Detection of Sensitive Information and Masking

- Spearheaded a cross-functional team of 8+ engineers as Founding AI engineer to architect an enterprise-grade Sensitive Information Discovery & Masking platform for global compliance (GDPR/PCI/PHI)
- Engineered a Hybrid PII/PHI Detection Engine using Spacy NER, Dependency Parsing, and heuristic Regex; achieved high-precision detection scores of 91% (PCI) and 87% (PII) across multi-modal data formats

- Developed a Multi-modal Extraction Pipeline utilizing PaddleOCR for images, PyMuPDF for PDFs, and Python-docx, ensuring 100% document format integrity during sensitive data obfuscation
- Built a High-Throughput Distributed System using Celery, RabbitMQ, and Redis to process and mask over 100GB of unstructured data in <2 hours across multi-cloud storage (S3, Azure, GCP)
- Architected a Scalable FastAPI Backend integrated with MongoDB and Rclone connectors, providing real-time UI reporting on folder-level sensitivity and automated data ingestion
- Managed the AI Lifecycle via MLFlow for model versioning and Docker for multi-container orchestration, ensuring a resilient, self-healing production environment

TheSmartCube

Nov 2021 – Apr 2023

Legal Clause & Commodity Extraction

- Developed a NLU Solution to automate the identification of prohibited commodities and exclusion clauses in legal PDFs; implemented RoBERTa classification with a custom Masked Language Model (MLM) for domain-specific tokenization
- Optimized OCR Accuracy by building a custom OpenCV solution to remove strike-through artifacts from scanned images prior to AWS Textract processing
- Deployed Serverless Inference using AWS Lambda and ECS, hosting XGBoost and Spacy NER models for real-time API-based clause identification

Multilingual Financial Data Intelligence

- Architected a Financial Extraction Engine to analyze US and German annual reports (Balance Sheets, Cash Flows); utilized Transformers for Machine Translation to normalize German data for English-based analysis
- Implemented XGBoost-based Table Localization to automatically identify page numbers and extract structured data via Tabula, reducing manual data entry by 80%
- Developed a Comprehensive API using Django to serve processed financial insights to front-end dashboards for global investment clients

Accenture

July 2019 – Nov 2021

Smart Search Engine using NLP

- Engineered a multi-format NLP search engine (PDF, DOC, PPT) utilizing TF-IDF and Cosine Similarity for low-latency semantic retrieval and optimized query relevance

SELECTED PROJECTS

- **AI-Powered Code Auto-Fixer** : Developed an LLM-based Autonomous Debugging System to detect, classify, and resolve codebase errors automatically; reduced average developer debugging time by 45%
- **LLM Reliability Router & Cost Tracker**: Architected a middleware solution for LLM orchestration featuring automatic failover, semantic caching, and real-time cost tracking

ACHIEVEMENTS

- Achieved 2nd place in the Machine Learning Hackathon at The Smart Cube
- Secured a top 10 ranking (top 0.003%) in the Global Technology Innovation Contest 2020

EDUCATION

Bachelor of Engineering in Electronics and Telecommunication | MITS, Gwalior | 2019

CERTIFICATIONS

- [TensorFlow Developer by deeplearning.ai](#) , [Generative AI Program - Google](#), [AWS Certified Machine Learning Specialty](#)